

PENUMALLU SYAM PRASAD REDDY

Software Development Engineer — AI/LLM Systems & Full-Stack Engineering

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Software Development Engineer with 2+ years of experience across the full stack, now focused on AI/LLM systems as a Senior AI Engineer at Beacon.li. Skilled in building scalable, high-performance web applications and in designing agentic and retrieval-augmented systems (RAG, LLM orchestration, semantic search) on modern cloud infrastructure. Comfortable owning a feature end-to-end — from system design through implementation, deployment, and monitoring — with a track record of mentoring and cross-functional collaboration.

CORE SKILLS

AI / LLM: RAG, Agentic Systems, LLM Orchestration, Semantic Search, Embeddings, MCP (Model Context Protocol), Vector Databases, Knowledge Graphs, IR Systems

Backend: Java, Spring, Spring Boot, Node.js, NestJS, Microservices, Event-Driven Architecture, ETL Pipelines

Frontend: Next.js, React, TypeScript, Micro-frontends, HTML, CSS, JavaScript, Bootstrap, Performance Optimization

Data & Cloud: MySQL, JDBC, AWS, GCP, Docker, Kubernetes, CI/CD, Observability

Leadership & Practice: System Design, Architecture, Mentoring, Tech Strategy, Problem-Solving, Time Management, Communication, Collaboration

Languages: Telugu (Native), English (Professional)

PROFESSIONAL EXPERIENCE

Senior AI Engineer — Beacon.li Apr 2026 – Present

- Design and build AI/LLM-powered systems, including retrieval-augmented generation (RAG) pipelines, agentic workflows, and semantic search over enterprise data.
- Work with LLM orchestration frameworks, vector databases, and knowledge graphs to improve retrieval accuracy and response quality.
- Partner with product and engineering stakeholders to translate business requirements into scalable AI system architecture.
- Mentor engineers on LLM application patterns and best practices for production-grade AI systems.

SDE 1 — Beacon.li Aug 2024 – Apr 2026

- Built and maintained full-stack features using Next.js, React, TypeScript on the frontend and Node.js, NestJS, Java/Spring Boot on the backend.
- Contributed to microservices and event-driven services, collaborating on system design and API contracts across teams.
- Deployed and monitored services on AWS/GCP using Docker and CI/CD pipelines, with a focus on observability and reliability.
- Worked on performance optimization for high-traffic frontend surfaces, including micro-frontend architecture.

Intern — Beacon.li May 2024 – Jul 2024

- Ramped up on the production codebase, contributing bug fixes and small features under senior engineer guidance.
- Gained hands-on exposure to the team's full-stack tooling, code review process, and deployment workflow.

EDUCATION

B.Tech, Electronics and Communication Engineering — JNTU Kakinada 2019 – 2023

Intermediate (MPC) — Sri Chaitanya College of Education 2017 – 2019

Secondary School (10th Grade) — Surya Public School, Drakshramam 2016 – 2017

CERTIFICATIONS

- Claude Certified Architect – Foundations
- Wells Fargo – Software Engineering Job Simulation
- AWS Cloud Virtual Internship — deployed serverless applications with AWS Lambda, API Gateway, and DynamoDB; built web applications using EC2, RDS, and S3.
- FLM Java Programming — built a CMS with Spring Boot, Hibernate, and MySQL; built a microservices application using Spring Cloud.
- CODESOFT Java Programming — developed enterprise applications with Java EE (EJB, JMS, JPA).